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IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc Communications and Signal Processing

DIGITAL IMAGE PROCESSING

Wednesday, 7 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer THREE questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Stathaki
Second Marker(s) : S. Parbhoo

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1. (a) Consider an $M \times N -$ pixel image $f(x,y)$ which is zero outside $0 \le x \le M-1$ and $0 \le y \le N-1$. Let $F(u,v)$ denote the $M \times N -$ point Discrete Fourier Transform (DFT) of $f(x,y)$. For compression purposes, we aim to discard transform coefficients with small magnitudes and retain only those with large magnitudes. Let $G(u,v)$ denote a modified version of $F(u,v)$, defined as

$$G(u,v) = \begin{cases} F(u,v), & \text{when } |F(u,v)| \text{ is large} \\ 0, & \text{otherwise} \end{cases}$$

We know that

$$\frac{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}|G(u,v)|^2}{\sum_{u=0}^{M-1}\sum_{v=0}^{N-1}|F(u,v)|^2} = \frac{19}{20}$$

We reconstruct an image $g(x,y)$ by computing the $M \times N -$ point inverse DFT of $G(u,v)$. Express

$$\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}(f(x,y)-g(x,y))^2$$

in terms of

$$\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}f(x,y)^2$$

[5]

(b) Consider an $M \times N -$ pixel real image $f(x,y)$, which is zero outside $0 \le x \le M-1$ and $0 \le y \le N-1$, with Discrete Fourier Transform (DFT) $F(u,v)$. A new DFT is created as $G(u,v) = F(u,v)$, followed by $G(u_0,v_0) = G(M-u_0, N-v_0) = 0$, where (u_0,v_0) is a specific two-dimensional frequency pair. A new image $g(x,y)$ is created by taking the Inverse Discrete Fourier Transform (IDFT) of $G(u,v)$. Find the difference $f(x,y) - g(x,y)$. **[5]**

(c) Consider the population of vectors $\underline{f}$ of the form

$$\underline{f} = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ f_3(x,y) \end{bmatrix}.$$

Each component $f_i(x,y), i = 1,2,3$ represents an image. The population arises from the formation of the vectors across the entire collection of pixels.
Consider now a population of vectors $\underline{g}$ of the form

$$\underline{g} = \begin{bmatrix} g_1(x,y) \\ g_2(x,y) \\ g_3(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve transforms of the vectors $\underline{f}$.
The covariance matrix of the population $\underline{f}$ calculated as part of the transform is

$$\underline{C_f} = \begin{bmatrix} a & b & 0 \\ b & a & 0 \\ 0 & 0 & c \end{bmatrix}$$

You can assume that b is a positive real number.

(i) Suppose that a credible job could be done of reconstructing approximations to the three original images by using one principal component image. What would be the mean squared error incurred in doing so, if it is known that $c < a - b$? **[5]**

(ii) Suppose that a credible job could be done of reconstructing approximations to the three original images by using two principal component images. What would be the mean squared error incurred in doing so, if it is known that $c > a + b$? **[5]**

2. (a) Consider an $M \times N -$ pixel image $f(x, y)$ which is zero outside $0 \leq x \leq M - 1$ and $0 \leq y \leq N - 1$. The histogram of $f(x, y)$ is sketched in **Figure 2.1** below.

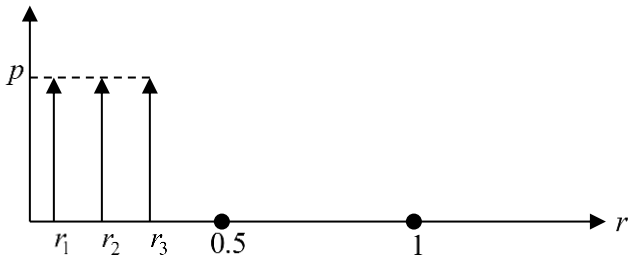


Figure 2.1

(i) Propose an intensity transformation function which will improve the contrast of the image when it is used to modify the intensity of the image. Sketch the histogram of the transformed intensity. **[5]**

(ii) Calculate the mean and the variance of the two images. **[5]**

(b) Propose a method for spatially adaptive noise reduction in images, that exploits jointly the following two observations:

1. Noise is typically less perceptible to human viewers in image regions of high visual detail, than in image regions of low detail.
2. Noise is also typically less perceptible to human viewers in bright image regions than in darker ones.

[5]

(c) We aim to apply a local threshold to the image $f(x, y)$ in order to identify pixels which are significantly brighter than their surroundings. We create an output $g(x, y)$ such that

$$g(x,y) = \begin{cases} 1, & f(x,y) > mean\{f(x,y)\} + T \\ 0, & \text{otherwise} \end{cases}$$

where $mean\{f(x, y)\}$ denotes the local mean computed over a 5×5 pixel neighborhood, and T is a user-defined threshold parameter. Explain how this process can be implemented using standard linear convolution followed by thresholding. What convolution kernel should be used to compute the local mean? **[5]**

3. We are given the blurred and noisy version $g(x,y)$ of an image $f(x,y)$ such that in lexicographic ordering

$$g = Hf + n$$

where H is the degradation matrix which is assumed to be block-circulant, and n is the noise term, which is assumed to be zero mean, independent from the original image and white. Moreover, f and g are the lexicographically ordered original and degraded image respectively.

(a) (i) Consider the Wiener filtering image restoration technique. Write down without proof the expressions for both the Wiener filter estimator and the restored image both in the spatial domain and the frequency domain. Explain all symbols used. **[5]**

(ii) Discuss the relation between Wiener filtering and both inverse and pseudo-inverse filtering. **[5]**

(b) Let $g(x,y)$ be a degraded, only by noise, image that can be expressed as

$$g(x,y) = f(x,y) + n(x,y)$$

where $f(x,y)$ is the original image and $n(x,y)$ is the background noise. In Wiener filtering we assume that the Discrete Space Fourier Transform $S_{ff}(u,v)$ of the autocorrelation function of the original image is available.

One method of estimating $S_{ff}(u,v)$ is to model the autocorrelation function as follows

$$R_{ff}(k,l) = \underset{(x,y)}{E}\{f(x,y)f(x+k,y+l\} = \rho^{|k|+|l|}$$

with ρ being an unknown parameter with $0 < \rho < 1$.

(i) Assuming $n(x,y)$ is a zero mean, white noise signal with unknown variance σ_n^2 and statistically independent of $f(x,y)$, write down without proof the expressions for the Wiener filter estimator and the restored image in the frequency domain as functions of ρ and σ_n^2. **[5]**

(ii) Develop a method to estimate ρ using samples of the autocorrelation function of the observed image $g(x,y)$. **[5]**

Hints:

1. The Discrete Space Fourier Transform $S_{ff}(u,v)$ of the autocorrelation function is defined as $S_{ff}(u,v) = \sum_{k=-\infty}^{\infty}\sum_{l=-\infty}^{\infty} R_{ff}(k,l)e^{-j(uk+vl)}$.
2. The following result holds: $\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}, |a| < 1$.

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4. (a) Consider an image with intensity $f(x,y)$ that can be modeled as a sample obtained from the probability density function sketched in **Figure 4.1** below:

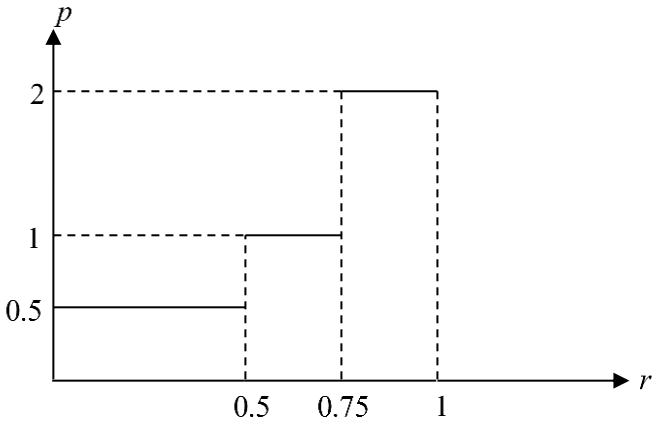


Figure 4.1

(i) Suppose that four reconstruction levels are assigned to quantize the intensity $f(x,y)$. Determine these reconstruction levels using a uniform quantizer. **[4]**

(ii) Explain briefly why uniform quantization may not be optimal in terms of minimizing the mean squared error. **[4]**

(iii) Determine the codeword to be assigned to each of the four reconstruction levels using Huffman coding. Specify the reconstruction level corresponding to each codeword. Based on your codeword assignment, calculate the average number of bits required to represent the quantized values. **[4]**

(iv) Compute the entropy, the redundancy and the coding efficiency of the Huffman code derived in part (iii). **[4]**

(b) Consider an image with intensity $f(x,y)$ that can be modeled as a sample obtained from the probability density function sketched in **Figure 4.2** below:

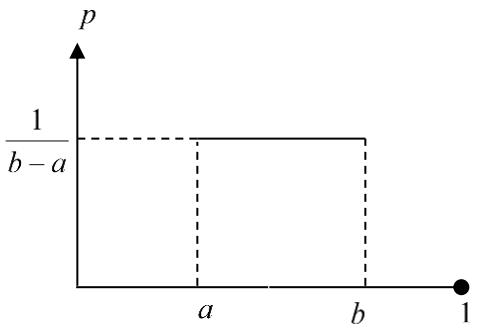


Figure 4.2

From $f(x,y)$ create an image

$$g(x,y) = f(x,y) - f(x-1, y-1).$$

Explain why applying symbol encoding to $g(x,y)$ is more efficient than applying symbol encoding directly to the original image $f(x,y)$. In your explanation, make use of the fact that adding uncorrelated images results in the convolution of their histograms**.** **[4]**